

# MEMORANDUM

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**To:** APSC 100 M3 Selection Team

**From:** Group L11, APSC 100, Tyce Renneberg, Keene Golden, Tina Song, Darwyn Montreuil, Khang Vo, Ruoyu Zhang

**Date:** November 30, 2025

**Subject:** Recommendations for an Assistive Device to squeeze toothpaste tubes.

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## Introduction

We have designed an assistive device to help individuals with limited hand mobility and dexterity, like Ken Fraser, squeeze toothpaste tubes. Individuals with similar disabilities often rely on caregivers to execute such tasks, limiting their independence. However, occupational care is sometimes inaccessible, which can result in the neglect of dental health altogether. Our project aims to address this gap in access by developing a 3D printed device that enables users to squeeze toothpaste from tubes easily. A key factor in our design process was minimizing the complexity of the motions the device uses. We also considered operability with various toothpaste tube sizes and reducing the amount of material used. This memo outlines our design process, including design considerations, concepts, and evaluations, that informs our final recommendation.

## Design Needs and Target Design Specifications

The device will be used by Ken Fraser, which makes him our primary stakeholder. Other stakeholders include occupational therapist Emma Smith and the 3D printing company. The most prominent needs for Ken Fraser focus on functionality, ease of use with the overall goal of gaining back his independence. For Emma Smith, ease of use and cost take priority so the device is accessible for clients. The 3D printing company has a larger focus on material used. See Appendix A for more details.

We then created specific and measurable design specifications, based on these needs. Some requirements for the design included a maximum weight of 100g and maximum material price of \$10. Beyond these quantifiable limits, the device should try to minimize material used, and maximize the number of tube sizes it can accommodate. For more detail, see Appendix B.

## Concepts Generated

Preliminary sketches of concepts were developed using the C-sketch method. Each member was then assigned a sketch to turn into a virtual prototype using a CAD-modelling software. With this, the following concepts were generated. See Appendix C for images of each.

1. Crankshaft: Uses a gear to wind up the tube and squeeze the toothpaste.

2. Slot & Rail: Comes in two pieces, one with a slot to squeeze the toothpaste and the other as a guiding rail to aid in mobility.
3. Box Roller: Contains the tube in a box and uses a sliding mechanism to push a roller and flatten the tube.
4. Angle-Force Assist: When the tube is placed in the slot and an angular force is applied, toothpaste is forced out of the tube.
5. Toothpaste Elevator: Using a loaded spring mechanism, pressing the base will result in the roller squeezing the tube.
6. EasyPress Tube: By pressing the large top surface, a vertical plunger drives an internal roller that smoothly pushes toothpaste toward the nozzle with minimal effort.

## Concept Evaluation

To begin, each member pitched their prototype to the rest of the group to familiarize everyone with every concept. Concepts were then screened against the design requirements (Appendix B). The Angle-Force Assist concept was perceived to be too difficult to use, so we integrated it into the Slot & Rail design.

Our five designs were then scored against the following evaluation criteria using a weighted decision matrix shown in Appendix D: Firstly, the amount of material used should be minimized which includes considerations of mass, volume, cost of material, and stress and strain with repeated use. This criteria was given a weight of 55%. Secondly the device should accommodate common tube sizes (travel, standard and large), which weights 25%. Next, the number of parts, an indicator of complexity, must be minimised, counting for 10%. And lastly, the use of the device should be subtle, another indicator of complexity. Subtlety was not determined much of a priority due to Ken Fraser while it could be significant for other potential users, so it only weighs 10%.

Overall, the Slot & Rail scored the highest overall, closely followed by the Crankshaft. This result stayed consistent after conducting a sensitivity analysis with a 10% tolerance in our decided weightings. As the Slot & Rail ranked consistently higher across the criteria, it was selected as our final design recommendation.

## Conclusion and Recommendations

The evaluation process demonstrated that the Slot & Rail, proposed by teammate number 2 of 6, consistently performed well among our four evaluation criteria. Compared to the other designs, the Slot & Rail makes the most efficient usage of material in 3D printing, which includes making use of internal ribs to effectively lower the cost. Additionally, the Slot & Rail is made up of only two parts, both of which tailor to Ken's specific range of hand motion. The handle on each component is long and thin which allows Ken to grip it in his fist, after which the design can be operated with a simple push motion, while keeping the toothpaste tube secure (See Appendix D). We recommend the Slot & Rail as an assistive toothpaste squeezing device.

## Appendix A: Needs Analysis

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ken Fraser:                | <ol style="list-style-type: none"> <li>1) Adaptive device should be accessible without assistance to give sense of independence</li> <li>2) Relatively easy to handle, small and lightweight, mobile               <ol style="list-style-type: none"> <li>a) Fits in a toiletry bag perhaps</li> </ol> </li> <li>3) Simple to use</li> <li>4) Durable for many use cycles without breaking</li> <li>5) Fits more than just single brand / style of tube (generalized)</li> </ol> |
| Occupational<br>Therapist: | <ol style="list-style-type: none"> <li>1) Low cost</li> <li>2) Intuitive design so that clients can learn with little difficulty</li> </ol>                                                                                                                                                                                                                                                                                                                                      |
| 3d printing<br>company:    | <ol style="list-style-type: none"> <li>1) Meets basic printing requirements and standards so that production is possible.</li> <li>2) Cheap and recyclable material</li> <li>3) Thin walls in designs should be avoided</li> </ol>                                                                                                                                                                                                                                               |

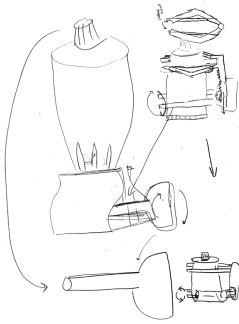
## Appendix B: Target Design Requirements and Evaluation Criteria

| R? | EC? | Target Design Specification                                                                                                                                        |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | ✓   | The mass of the device is no more than 100g (less is better).                                                                                                      |
| ✓  | ✓   | Cost is minimized, with a maximum price of \$10.                                                                                                                   |
| ✓  | ✓   | The device's max dimensions are no more than 20 cm x 10 cm x 10 cm (less is better).                                                                               |
| ✓  | ✓   | The device must work for at least these two standard sizes of tubes (more is preferred).<br>a. Travel size (100mL), standard size (170mL).                         |
| ✓  |     | Works with screw off and flip up lid style tubes.                                                                                                                  |
| ✓  | ✓   | The device can withstand usage two to three times a day for at least a year (at least 1000 dispense cycles).                                                       |
|    | ✓   | The design minimizes the use of moving parts.                                                                                                                      |
| ✓  |     | Must dispense consistent amounts of tube substance (pea sized = 0.25 g a serving = 0.2 cm <sup>3</sup> ).                                                          |
| ✓  |     | The device can be operated by one person, without assistance.                                                                                                      |
| ✓  |     | The device can be set up and used using i) between two clenched hands, ii) with one's mouth and teeth, iii) slotted in a clenched hand using force from a surface. |
| ✓  | ✓   | Components can be physically assembled and produced using nylon PLA SLS printed parts (using fewer individually printed parts is better).                          |
|    | ✓   | The aesthetics of the device are age appropriate.                                                                                                                  |
| ✓  |     | All corners are rounded.                                                                                                                                           |
|    | ✓   | The product's ergonomics increase user comfort during use (minimizes muscle strain, minimized forces, shape).                                                      |
|    | ✓   | Ability to fit a standard cap size, and twist it off, and re-attach.                                                                                               |

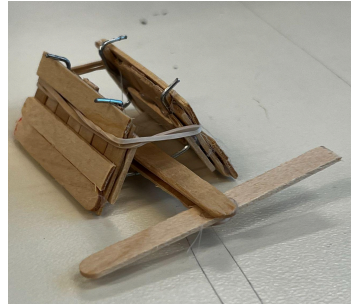
## Appendix C: Prototypes

### 1. Crankshaft

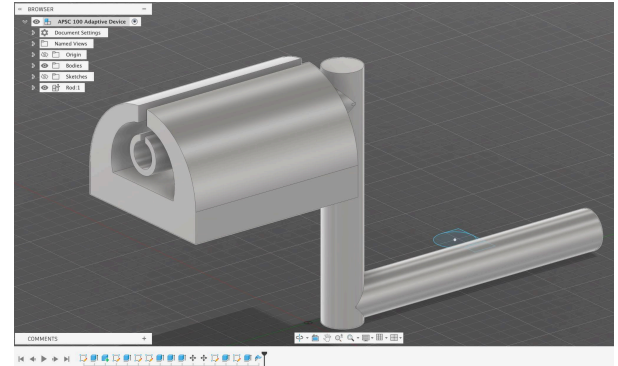
Sketched and prototyped by member 1 of 6. The design features a base in which the empty tube can be coiled, and a rotating piece to coil the tube.



**Figure 1: Initial C-Sketch**



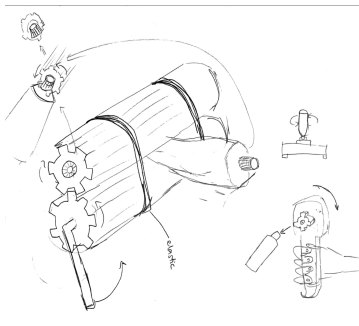
**Figure 2: Physical Prototype**



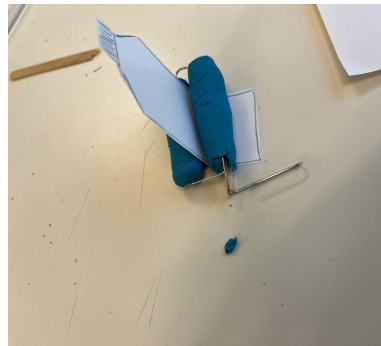
**Figure 3: CAD Model with design tree**

### 2. Slot & Rail

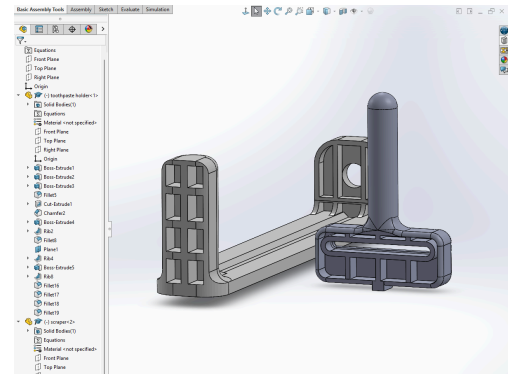
Sketched and prototyped by member 2 of 6. The design originally planned to use two interlocked gears to force toothpaste out of the tube. The design went through multiple iterations between the physical and CAD stages to simplify the mechanism and reduce complexity.



**Figure 4: Sketch of Slot & Rail**



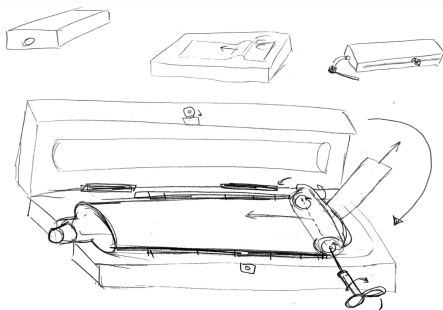
**Figure 5: Physical Prototype**



**Figure 6: CAD model with design tree**

### 3. Box Roller

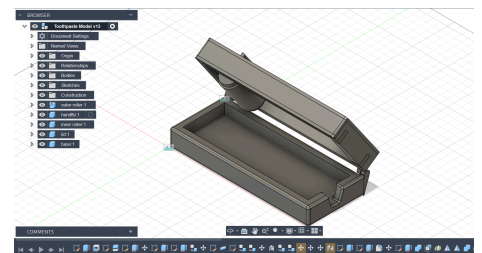
Sketched and prototyped by member 3 of 6. This design features a hinged box with a roller attached to the lid that can be pushed to slide across the box and dispense toothpaste.



**Figure 7: Initial C-Sketch**



**Figure 8: Physical Prototype**



**Figure 9: CAD Model**

#### 4. Angle-Force Assist

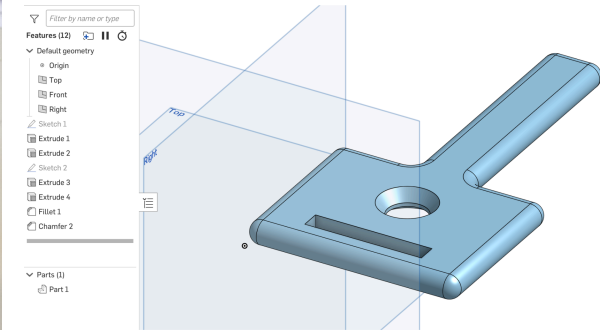
Sketched and prototyped by member 4 of 6. The primary idea of the design focuses on using an angled-slot that the user can leverage against to force toothpaste out of the tube.



**Figure 10: Initial C-Sketch**



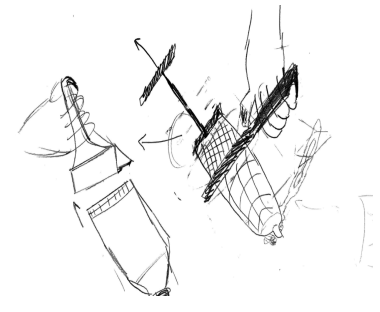
**Figure 11: Physical Prototype**



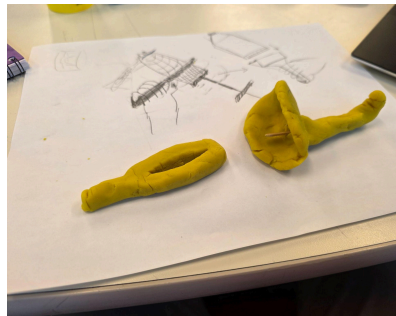
**Figure 12: CAD Model with design tree**

#### 5. Toothpaste Elevator

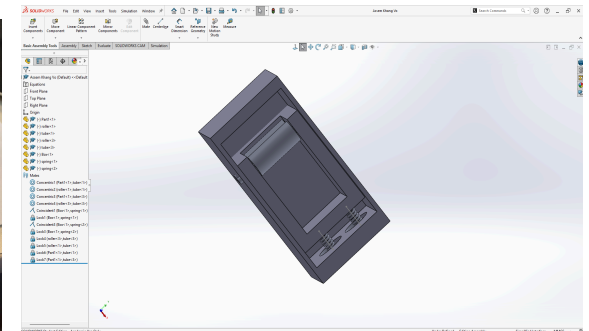
Sketched by member 5 of 6. The sketch (Figure 13) featured a slot and stationary clip to pull against. During the CAD stage, the design evolved to simplify the motions. Now, it lets the user clip in a tube and press the spring-loaded plate with a toothbrush, squeezing the toothpaste out.



**Figure 13: Initial C-Sketch**



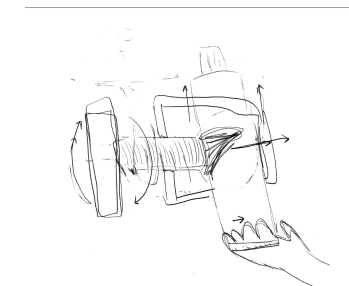
**Figure 14: Physical Prototype**



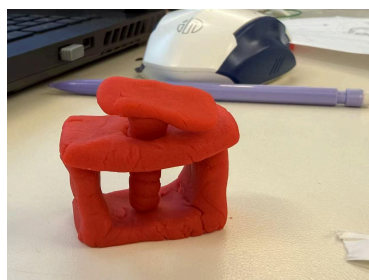
**Figure 15: CAD Model with design tree**

#### 6. EasyPress Tube

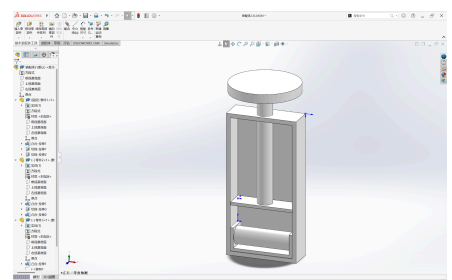
Initially sketched and prototyped by member 6 of 6. The EasyPress Tube is palm-activated. A vertical plunger moves downward with a simple press. A roller then pushes toothpaste forward. The device requires minimal hand strength and improves accessibility.



**Figure 16: Initial C-Sketch**



**Figure 17: Physical Prototype**



**Figure 18: CAD Model with design tree**

## Appendix D: Weighted Decision Matrix

| Evaluation Criteria     | Weight | Crankshaft |                | Slot & Rail |                | Toothpaste Elevator |                | EasyPress Tube |                | Box Roller |                |
|-------------------------|--------|------------|----------------|-------------|----------------|---------------------|----------------|----------------|----------------|------------|----------------|
|                         |        | Score      | Weighted Score | Score       | Weighted Score | Score               | Weighted Score | Score          | Weighted Score | Score      | Weighted Score |
| Material used           | 55%    | 9.09       | 5.00           | 10.00       | 5.50           | 0.00                | 0.00           | 6.00           | 3.30           | 0.00       | 0.00           |
| Subtlety                | 10%    | 7.66       | 0.77           | 9.45        | 0.94           | 0.00                | 0.00           | 10.00          | 1.00           | 7.77       | 0.78           |
| Tube Size Accommodation | 25%    | 10.00      | 2.50           | 9.00        | 2.25           | 9.00                | 2.25           | 10.00          | 2.50           | 9.00       | 2.25           |
| Number of Parts         | 10%    | 8.00       | 0.80           | 8.00        | 0.80           | 0.00                | 0.00           | 6.00           | 0.60           | 6.00       | 0.60           |
| Total                   | 100%   |            | 9.07           |             | 9.49           |                     | 2.25           |                | 7.40           |            | 3.63           |

### Justification for Weightings and Scoring

**Material Used:** Accounts for cost, mass, volume of material used. It indicates the ease of use and comfort for the user. Therefore, it should have a strong weight because it accounts for many needs identified. Measured using volume of material (from CAD design).

**Subtlety:** Determines how portable and large the device is. Fraser doesn't travel often and seems to be publicly accepting of his disability, so this criteria is less important. Measured from dimension of device (from CAD model).

**Tube Size Accommodation:** With a range of toothpaste tube sizes, it is preferred to accommodate for more. Measured from the length and width of tube size range which can fit in the design.

**Number of Parts:** More moving parts typically means a less durable, more complicated product. So, attempting to minimize these is important for manufacturing and sustainability but less so for Ken directly.

## Appendix E: Gen AI Usage Statement

"No GenAI tools were used in the preparation of this memo or any of its contents."